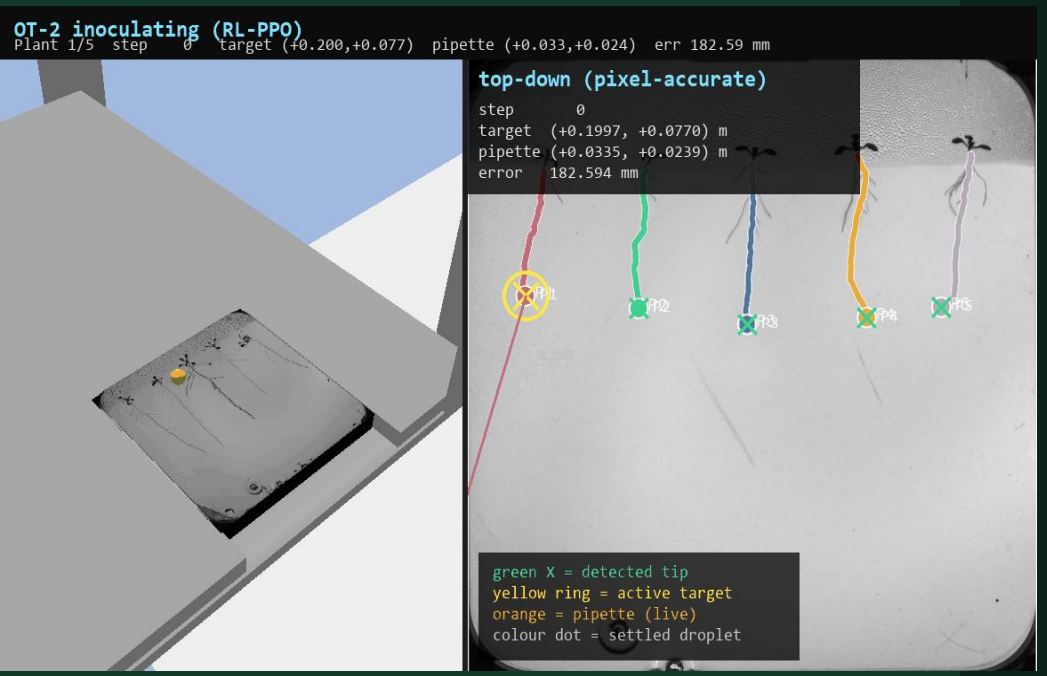


BLOCK B · CREATIVE BRIEF · RETAKE

From Petri Dish to Pipette

Automated root phenotyping and robotic inoculation, end to end

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THE PROBLEM · ILO 5.1

Measuring and treating roots, one seedling at a time

Plant scientists grow seedlings on agar plates, study how the roots develop, then inoculate each root tip with bacteria or compounds, for example to test which genotypes resist disease.

Both steps are done by hand today: measuring root length off photos, and pipetting onto each tiny tip under a microscope.

It is slow, tiring and hard to reproduce, which caps how many plants and conditions a lab can ever screen.

slow

minutes of microscope work per plant

variable

hand placement drifts run to run

doesn't scale

to the hundreds of plants needed

WHY IT HELPS THE CLIENT · ILO 5.1

What automation unlocks for plant science

Screen far more plants

Photograph a plate and find every root automatically — throughput jumps from a handful to hundreds per session.

Spot resistant genotypes

More plants measured means cleaner comparisons between lines, so disease-resistant candidates surface sooner.

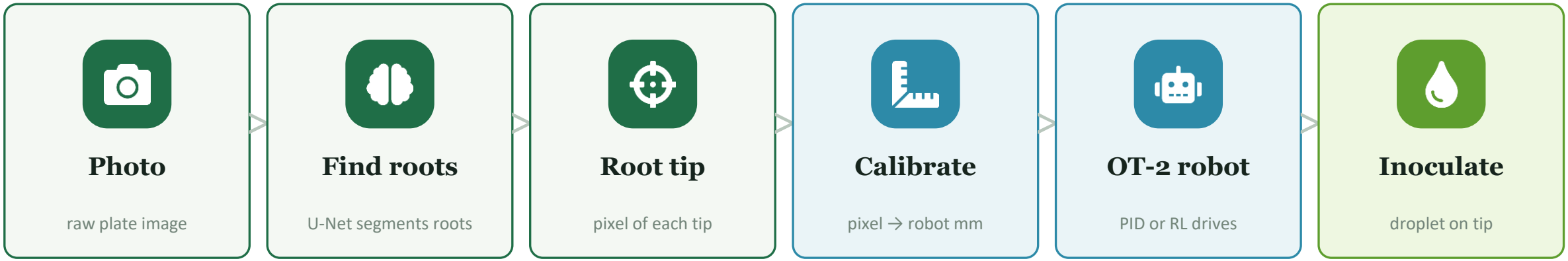
Reproducible inoculation

The robot dispenses on the same spot every time, removing the hand-to-hand variation that muddies results.

In short: one scientist can do in an afternoon what used to take a week of microscope time.

THE SOLUTION · ILO 8

One pipeline: photo in, inoculated root out

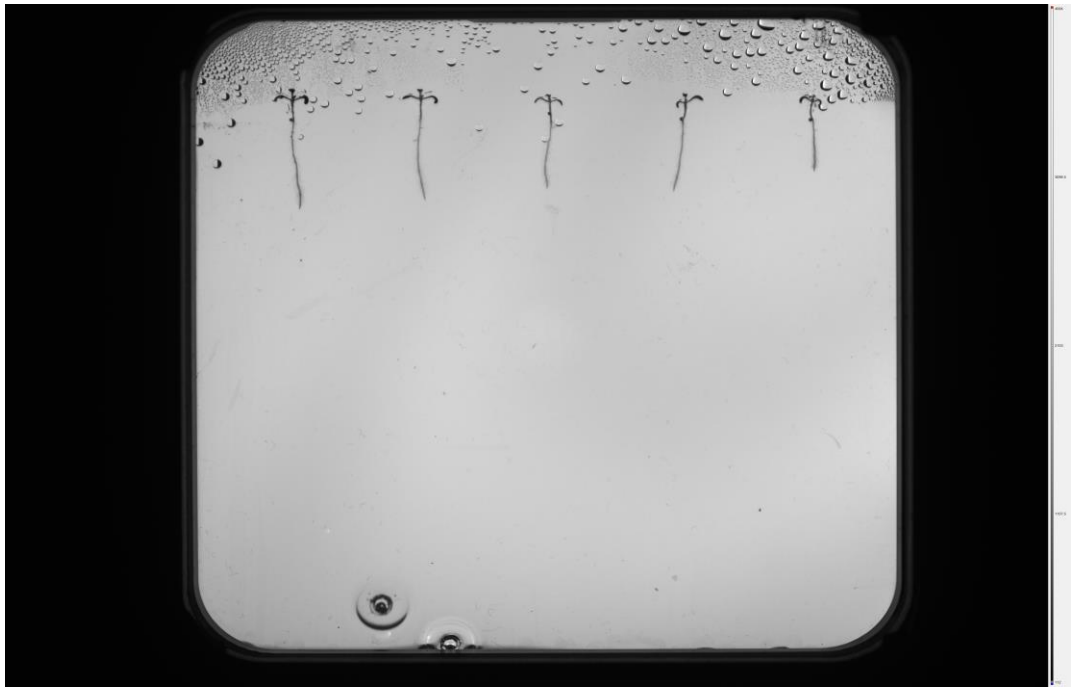


Two halves: **computer vision** finds the targets, **robotics** hits them. Calibration is the bridge between a pixel and a millimetre.

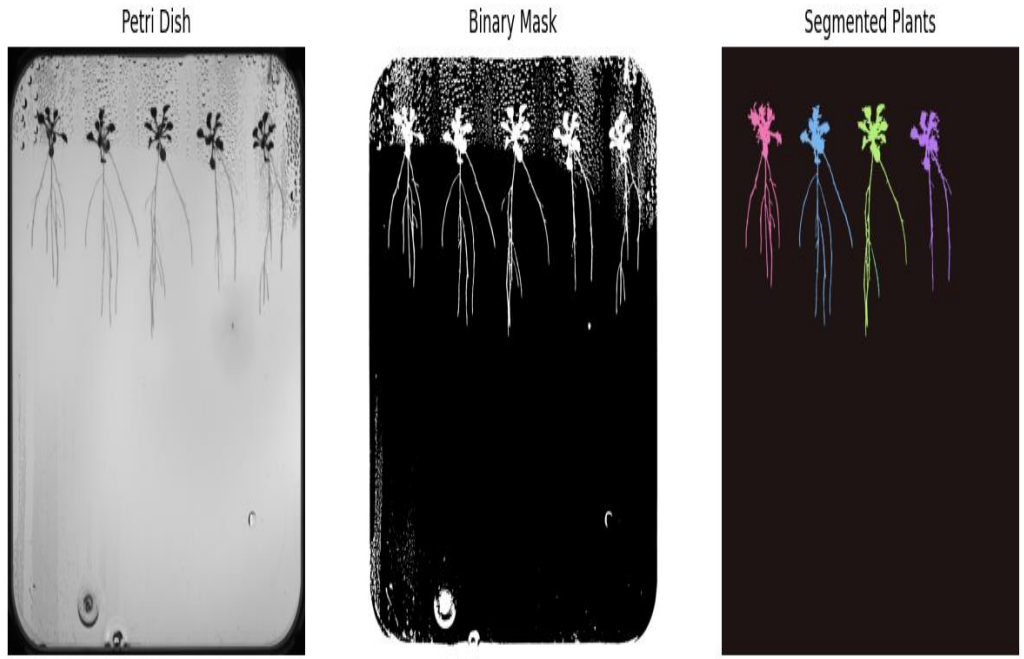
COMPUTER VISION · ILO 6.4 · TASKS 2–3

Traditional CV: crop the dish, separate the plants

Classic image processing first, as the brief asks — enough to isolate the working area before any model runs.



Task 2 — petri dish cropped from the raw image (threshold + contours)



Task 3 — the five plants separated as instances, no deep learning

COMPUTER VISION · ILO 8.5 · TASKS 5 & 8

Deep learning: a U-Net that segments roots



Task 5 — U-Net root mask predicted on a test plate

F1 = 0.877

validation score of the root U-Net (Task 5)

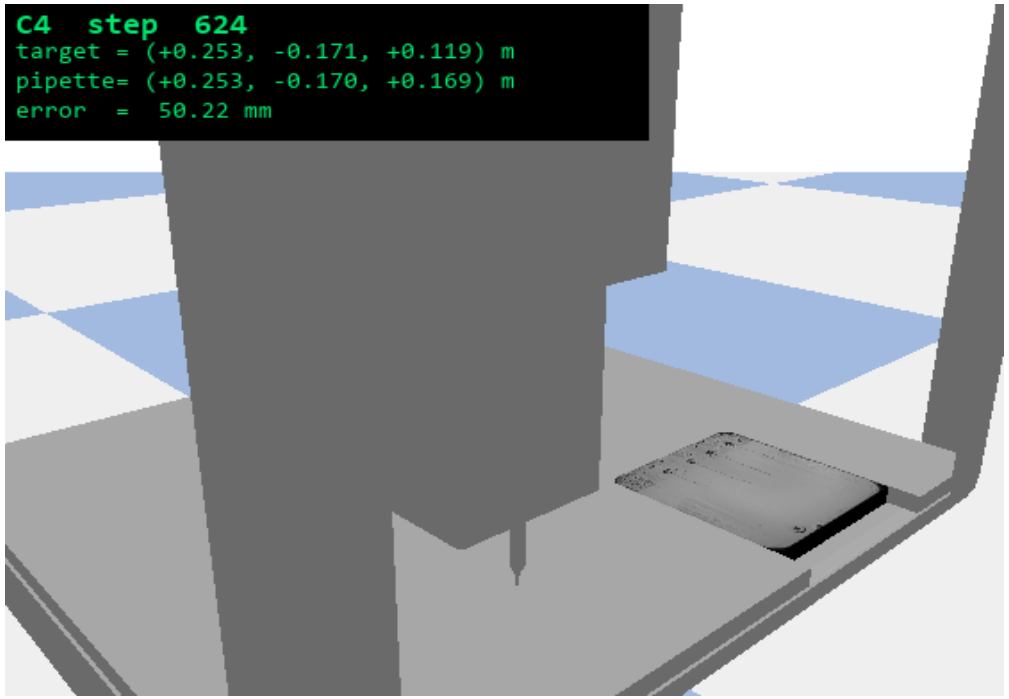
Task 8 — full Kaggle pipeline:

dish crop → U-Net → skeletonise the root system → measure primary-root length for every plant.

These tip coordinates feed straight into the robot.

ROBOTICS · ILO 8.6A · TASK 9

Meet the robot and learn its reach



Driving the pipette to the corners of the workspace

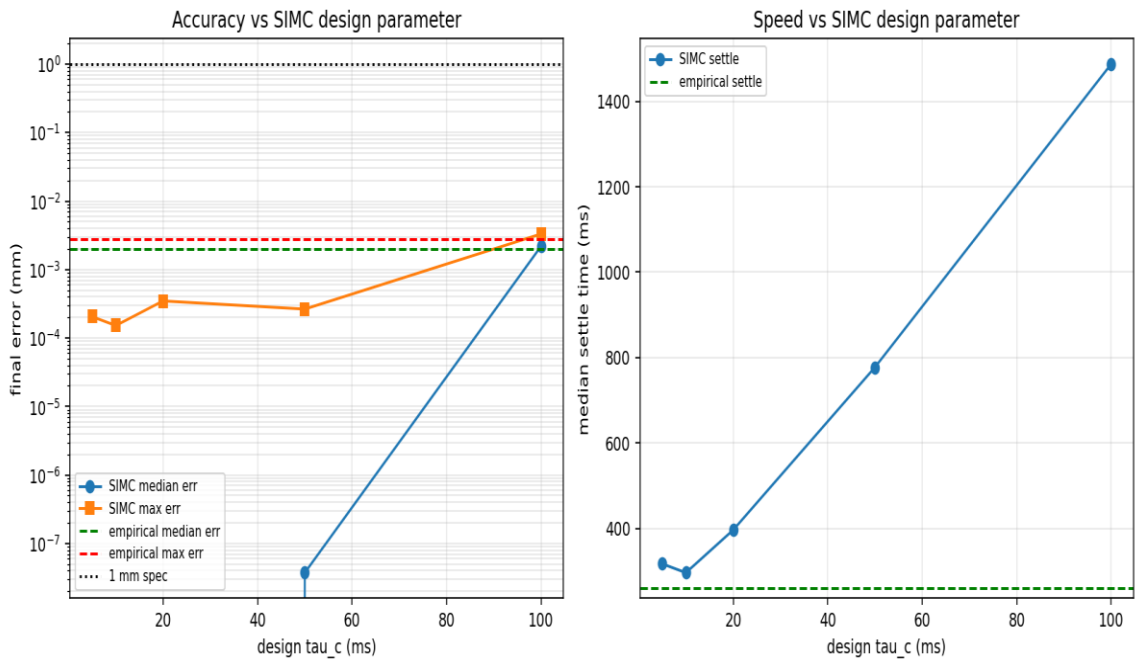
The OT-2 is a real 3-axis pipetting robot (here, its digital twin). Before controlling it, I mapped what it can actually reach.

Key finding: only 5 of the 8 workspace corners are reachable — the chassis floor blocks the lowest layer.

So every later task targets a practical envelope, not the full box. Saved a lot of failed runs.

ROBOTICS · ILO 8.6B · TASK 10

Classic control: a PID tuned from physics



Step-response fit (system ID) → SIMC tuning rule → PID gains

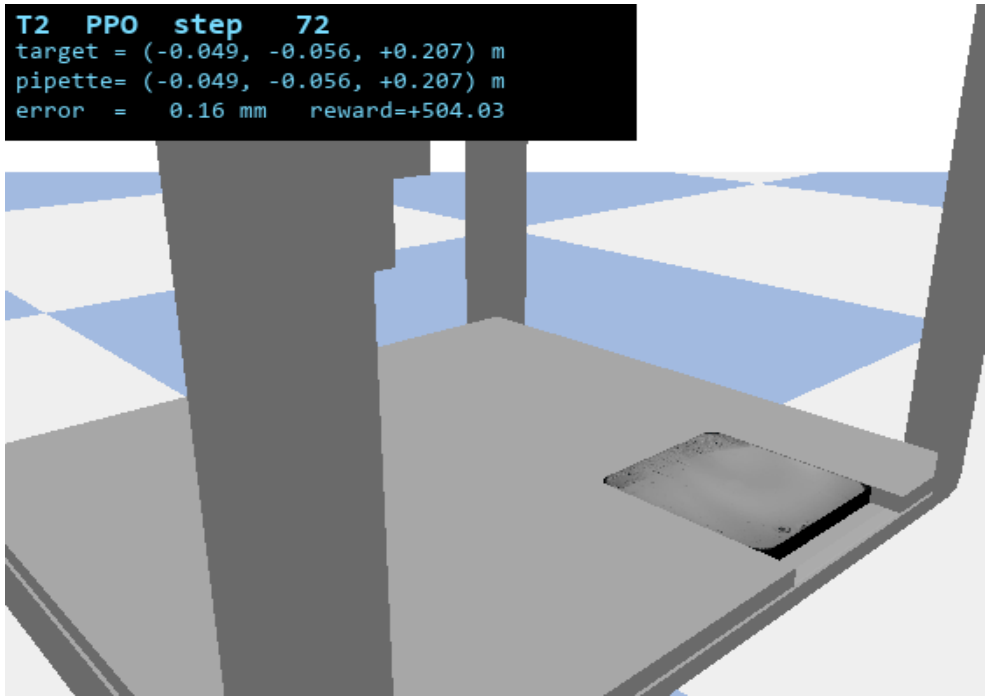
Instead of guessing gains, I measured how each axis responds, fit a first-order model, and computed the PID directly with the SIMC rule.

sub-micron

worst-case error in simulation (0.3 µm) — far inside the 1 mm spec

ROBOTICS · ILO 8.6C · TASK 11

Reinforcement learning: a policy that learns to aim



PPO policy steering the pipette to a target

A PPO agent learns control purely from trial and reward — no physics model handed to it.

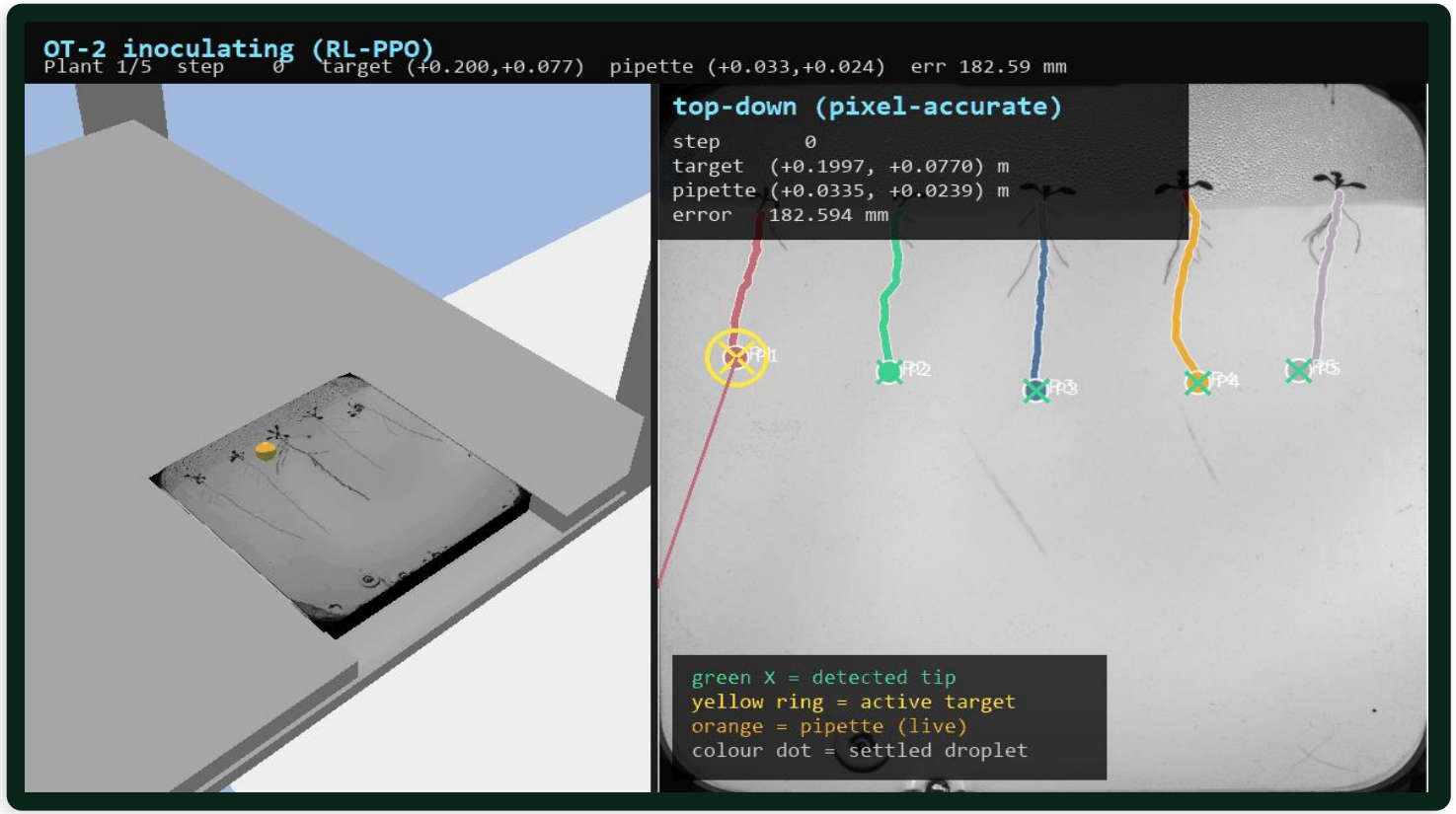
The reward design was the whole game: three proximity wells at 10 mm, 0.3 mm and 0.05 mm pull it from coarse approach down to sub-millimetre.

60 / 60

random targets hit under 1 mm (trained policy)

INTEGRATION · ILO 8.6D · TASK 12

Putting it together: see a root, hit a root



▶ *plays live in slideshow*

The autonomous loop

- U-Net finds each primary-root tip
- calibration turns the tip pixel into robot mm
- the controller drives there and dispenses
- repeat for all five plants

RESULTS & EVALUATION · ILO 8.6D · TASK 13

PID vs RL across 19 dishes, 82 root tips

100%

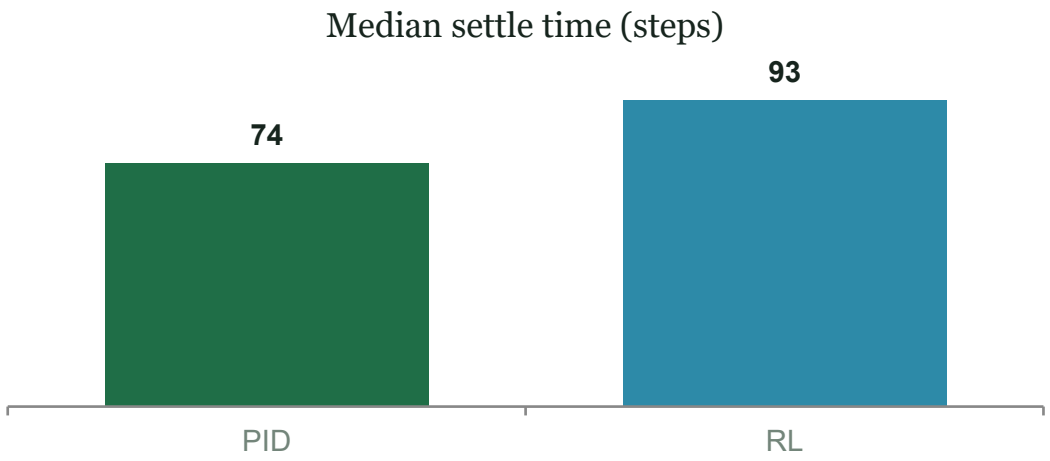
tips < 1 mm — both controllers

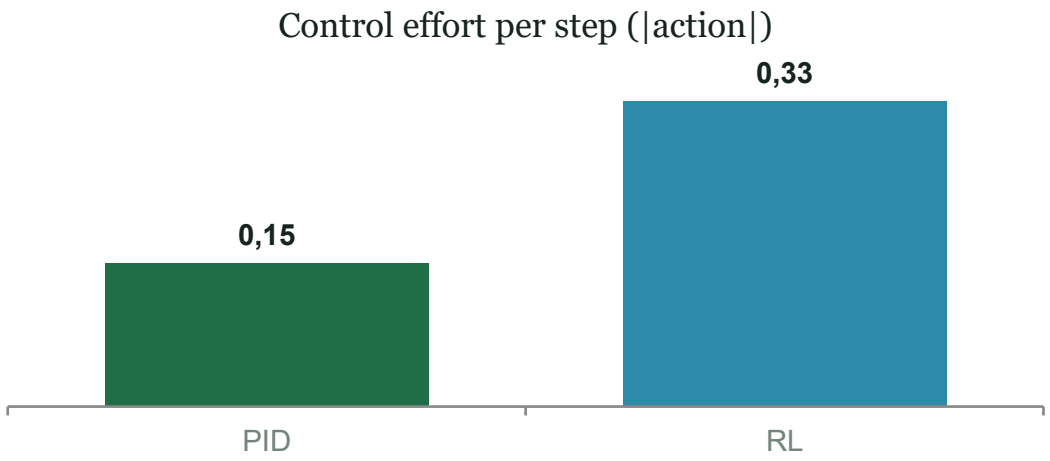
sub-µm

PID median error

0.47 mm

RL median error





Both meet the spec under a fair test (same start, same horizon). PID wins on raw precision because it has a plant model; RL needs no model and still clears 1 mm on every tip — at a bit more effort.

ERROR ANALYSIS & ITERATION · ILO 2.6

Three fixes that actually moved the numbers

For each: what broke, what I changed, and the before / after.

ASSUMPTIONS

What I took as given

Five plants per dish

Detection keeps the five largest root blobs, ordered left to right.

Known dish geometry

The plate is a fixed 150 mm square at a known spot on the deck.

Idealised specimen

In the sim the plant is a flat target — no occlusion, condensation or 3-D roots.

Hover-and-drop dispense

The pipette inoculates from just above the agar, not by touching it.

NEXT STEPS · ILO 8

Where I'd take it next

Add a shoot model

Segment shoots too, so plant localisation no longer depends on the root mask alone.

Hybrid controller

RL for the fast coarse approach, hand off to PID for the final sub-mm settle.

Tighter RL

Re-train against a 0.5 mm target; early signs suggest it can roughly halve its error.

Real hardware

Re-calibrate on a physical OT-2 and switch to a touch-dispense at the agar.

IN ONE LINE

A camera finds every root tip, and the robot inoculates all of them — autonomously, under a millimetre.

100%

tips hit < 1 mm (PID & RL, 82 tips)

0.877

U-Net root-segmentation F1

5 of 5

plants found per dish after the fix

Mohammadali Jaberi 244437 Thank you — questions welcome